



FOOD

ID	Indicator	Description
ID_FOOD_01	projected change of cereal yields	Projected amount that climate change is predicted to change food supply by mid-century for three staples: rice, wheat and maize. The projections of the yield productions are obtained from five crop models (EPIC, GEPIC, LPJmL, pDSSAT, PEGASUS), and it assumes effect of CO2 fertilization but does not adjust for changes in farming systems or irrigation.
ID_FOOD_02	projected population change	An indication of food demand by the mid-century. The projection data are from the World Bank Health Nutrition and Population Statistics (HNPStats) which provides country-level projection of population up to 2050.
ID_FOOD_03	food import dependency	Food comprises commodities such as food and live animals, beverages and tobacco, and animal and vegetable oils, such as fats and oil seeds, oil nuts, and oil kernels. This indicator is expressed as a percentage of merchandise imports which is comprised of goods whose economic ownership is changed between a non-resident and a resident and that are not included in the following specific categories: goods under merchanting, non-monetary gold, and parts of travel, construction, and government goods and services n.i.e.
ID_FOOD_04	rural population	The proportion of the total population living in rural areas, defined as the difference between total population and urban population according to national statistical offices.
ID_FOOD_05	agriculture capacity	A combination of four indicators of agricultural technology: capacity to equip agriculture areas with irrigation, N+P205 total fertilizer use on arable and permanent crop area use, pesticide use, and tractor use. The irrigation measure obtained from FAO indicates the proportion of agricultural areas equipped with irrigation, but does not measure the amount of land that has indeed been irrigated in a specific year. Therefore, it is a capacity measure. The fertilizer and pesticide measures are the total consumption of the active ingredients (for both fertilizer and pesticide) as the reported sum divided by hectare. The tractor use measures the number of wheel and crawler tractors used in agriculture. Together, these measures are combined into an indication of the accessibility of agricultural technological inputs.
ID_FOOD_06	child malnutrition	A measure of malnutrition based on the percent of under-5-year-olds with a low weight for height ratio; usually taken as a good indicator of chronic malnutrition. Generally, OECD countries are not reported, so ND-GAIN uses 0%.



WATER

ID	Indicator	Description
ID_WATE_01	projected change of annual runoff	An indication of how climate change will bring changes to annual surface water resources by the mid of the century. Projected surface runoff data, defined as precipitation minus evapotranspiration and change in soil moisture storage, are provided by Aqueduct at the World Resource Institute. Aqueduct uses the ensemble of six global circulation models (GCMs) from Coupled Model Intercomparison Project Phase 5 (CMIP5) chosen to represent a broad diversity of models that best reproduce the mean and standard deviation of recent streamflow records in 18 large river basins (Alkama et al., 2013). The database covers 14998 catchments derived from the Global Drainage and Basin Database.
ID_WATE_02	projected change of annual groundwater recharge	An indication of how climate change will bring changes on annual groundwater resources by mid-century. GWR data are provided by Goethe University Frankfurt (Portmann et al., 2013).
ID_WATE_03	fresh water withdrawal rate	The proportion of total actual renewable water resources that is withdrawn as freshwater, to approximate the pressure on the renewable water resources, according to the FAO Aquastat database.
ID_WATE_04	water dependency ratio	The proportion of the total renewable water resources originated outside the country, including the surface water and ground water entering the country or secured by treaties.
ID_WATE_05	dam capacity	An indication of the capacity to adjust to the changing (temporal and geographical) distribution of freshwater resources, including changes due to climate change. It is a measure of the per capita dam storage capacities within one country, calculated by the per capita theoretical initial capacities of all dams, which does not allow for changes over time due to siltation.
ID_WATE_06	access to reliable drinking water	Commonly used indicator of the capacity to deliver reliable domestic water supplies. The drinking water sources are considered reliable if they have a household connection, public standpipe, borehole, protected well or spring, or rainwater collection.



HEALTH

ID	Indicator	Description
ID_HEAL_01	projected change of deaths from climate change induced diseases	An indication of the impact of climate change on several types of diseases. The indicator is a model-based estimate of the quality-adjusted loss of life years under several different climate scenarios. Disability adjusted life year (DALY) due to malaria, an indication of the climate change impacts on vector borne diseases, is excluded because more specific models have been used to project such impacts and it is assessed by another ND-GAIN indicator, the projected change of length of transmission season of vector-borne diseases.
ID_HEAL_02	projected change in vector-borne diseases	This indicator takes the projection of malaria LTS as an indication of the climate change impacts on vector-borne diseases. LTS data were taken from projections (Caminade, et al., 2014) that took the ensemble mean of malaria LTS over four malaria models and five GCMs. However, the incidence of vector-borne diseases is also strongly dependent on the quality of public health systems. In this indicator the WHO estimated number of malaria cases per 1000 population per month of current LTS is used as a measure of these services.
ID_HEAL_03	dependency on external resource for health services	Share of current health expenditures funded from external sources. External sources compose of direct foreign transfers and foreign transfers distributed by the government encompassing all financial inflows into the national health system from outside the country.
ID_HEAL_04	slum population	The proportion of urban population living in slum households, defined as a group of individuals living under the same roof lacking one or more of life-supporting facilities: access to improved water, access to improved sanitation, sufficient-living area and durability of housing. Generally, OECD countries are not reported, so for this indicator ND-GAIN assumes 0%.
ID_HEAL_05	medical staff	Sum of the number of physicians, nurses and midwives per 1000 population in the country. Increases in physicians, nurses, or midwives will have the same effect on the indicator.
ID_HEAL_06	access to improved sanitation facilities	The percentage of people using improved sanitation facilities that are not shared with other households and where excreta are safely disposed of in situ or transported and treated offsite. Improved sanitation facilities include flush/pour flush to piped sewer systems, septic tanks or pit latrines: ventilated improved pit latrines, composting toilets or pit latrines with slabs.



ECOSYSTEM SERVICES

ID	Indicator	Description
ID_ECOS_01	projected change of biome distribution	An indication of how climate change will impact the change of terrestrial biome biodiversity within a country by the end of the century. Data were taken from the global version of a dynamic vegetation model (MC1) (Gonzalez et al., 2010).
ID_ECOS_02	projected change of marine biodiversity	An indication of how climate change will impact the change of marine biodiversity in a country's exclusive economic zones by mid-century. It is a measure based on projected changes in the distribution of 1066 exploited species of marine fish and invertebrates under climate envelope scenarios based on A1B scenarios (Cheung et al., 2009).
ID_ECOS_03	natural capital dependency	Based on the World Bank's Natural Capital Accounting project. This indicator of the strength of the dependency of social systems on ecosystem goods and services is based on the deployment of natural capital in national accounting, including national income and savings in the form of all assets and capital goods that are inputs to economic well-being (The World Bank, 2011). The natural capital related to ecosystem services includes: crop, pasture, forest (timber), forest (non-timber) and protected areas. Sub-surface capital such as oil, gas and mineral reserves are not Included.
ID_ECOS_04	ecological footprint	The ecological footprint estimates the number of hectares of land and water, both within and outside the country, that are needed to meet the average demand on ecosystem services by the population's lifestyle. This is compared with the estimated capacity of a country's ecosystems to regenerate and maintain ecosystem services for either internal use or export. This indicator uses the surplus or deficit of capacity to cover the demand within each country.
ID_ECOS_05	protected biomes	Taken directly from the Yale Environmental Performance Index (EPI), the indicator "assesses the protection of biomes weighted by the proportion of a country's territory the biome occupies." EPI defines the indicator as follows: "It measures the degree to which a country achieves the target of protecting 17% of each terrestrial biome within its borders, weighted by the domestic contribution of each terrestrial biome...all biome protection percentages were capped at 17% so that higher protection in one biome cannot be used to offset lower protection in another."
ID_ECOS_06	engagement in international environmental conventions	An indicator based on the country's participation in international forums, which is an indicator of its capacity to engage in multilateral negotiations and to reach agreement on appropriate actions internally.



HUMAN HABITAT

ID	Indicator	Description
ID_HABI_01	projected change of warm periods	An indication of the probability of extreme heat under climate change by mid-century. This indicator uses the Warm Spell Duration Index (WSDI), which defines periods of excessive warmth using a percentile-based threshold calculated for a calendar 5-day window in the base period 1961-1990. WSDI counts the number of days in a year when the daily maximum of near surface temperature exceeds the 90th percentile threshold for 6 consecutive days or longer (Alexander, et al., 2006; Sillmann, et al., 2013b).
ID_HABI_02	projected change of flood hazard	Flood hazard is measured by the predicted, monthly maximum precipitation in 5 consecutive days (rx5day). Rx5day is defined as monthly maximum consecutive 5-day precipitation. It is a measure of precipitation extreme under climate change, a risk factor for flood hazard (Kundzewicz & Schellnhuber, 2004). The monthly rx5day data are extracted from ensemble mean of extreme indices generated by 19 GCMs (Sillmann et al., 2013a; Sillmann et al., 2013b).
ID_HABI_03	urban concentration	Urban concentration measures both concentration of a country's population within cities (i.e., the degree of urbanization in general) and concentration of the urban population within a small number of large population (cities of 750,000 inhabitants or more) centers via the Herfindahl Index (Henderson, 2000; Van Eck & Koomen, 2008).
ID_HABI_04	age dependency ratio	An indication of the size of the vulnerable population in terms of ages. This indicator considers the population under 14 or above 65 as the vulnerable group.
ID_HABI_05	quality of trade and transport infrastructure	Logistics professionals' perception of a country's quality of trade and transport related infrastructure (e.g., ports, railroads, roads, information technology), on a rating ranging from 1 (very low) to 5 (very high). Scores are averaged across all respondents.
ID_HABI_06	paved roads	Roads surfaced with crushed stone (macadam) and hydrocarbon binder or bituminized agents, with concrete, or with cobblestones, as a percentage of all the country's roads, measured in length. It reflects a country's capacity to acquire and deploy transportation improvements, especially in rural areas.



INFRASTRUCTURE

ID	Indicator	Description
ID_INFR_01	projected change of hydropower generation capacity	An indication of the potential risk of hydropower generation capacity weighted by the importance of hydropower to one country, i.e., the proportion of the electricity production from hydroelectric sources. The data of the projected change are available at the sub-continental level, drawn from (Hamududu & Killingtveit, 2012).
ID_INFR_02	projected change of sea level rise impacts	An indication of how coastal infrastructure will be impacted by the combined effect of sea level rise and potential storm surge by the end of the century. The indicator considers the proportion of land areas, adjacent to the ocean, that are lower than the projected sea level rise and the average height of storm surge.
ID_INFR_03	dependency on imported energy	A measure of the percentage of total energy use that is imported and thus not fully within a country's control. According to the IEA, energy use refers to the use of primary energy before transformation to other end-use fuels, equal to indigenous production plus imports and stock changes, minus exports and fuels supplied to ships and aircraft engaged in international transport.
ID_INFR_04	population living under 5m above sea level	The proportion of the population living in the area where elevation is 5 m or less. It is a simple measure of the population sensitive to coastal risks.
ID_INFR_05	electricity access	The proportion of the population with access to grid-power. Generally, OECD countries are not reported, so for this indicator ND-GAIN assumes a rate of 100%.
ID_INFR_06	disaster preparedness	A measure of a nation's adoption and implementation of national disaster risk reduction strategies in line with the Sendai Framework for Disaster Risk Reduction 2015-2030, as measured through the UN Sustainable Development Goal 13.1.2.

ECONOMIC READINESS

ID	Indicator	Description
ID_ECON_01	doing business (retired in 2022)	The indicator took the World Bank Doing Business (DB) indicators as an indication of how countries are capable of attracting adaptation investment. The index assesses the investment climate in 10 topics using 40 indicators. The 10 topics are: starting a business, dealing with construction permits, getting electricity, registering property, getting credit, 35 protecting investors, paying taxes, trading across borders, enforcing contracts, and resolving insolvency.
ID_ECON_02	Business ready (integrated in 2026, backfilled to 2022)	The indicator uses the World Bank Business Ready (B-Ready) Index, for the countries currently covered by its staged release, as an indication of the capability of a country to attract adaptation investment. The B-Ready Index assesses the investment climate across three pillars (regulatory framework, public services, and efficiency) and 10 topics: Business Entry, Business Location, Utility Services, Labor, Financial Services, International Trade, Taxation, Dispute Resolution, Market Competition, and Business Insolvency. Across the indicators, the regulatory framework is captured in policy and law, while public services and efficiency is generally measured through primary data collection.

GOVERNANCE READINESS

ID	Indicator	Description
ID_GOVE_01	political stability and non-violence	An indicator directly from the World Governance Indicators (WGI), “capturing perceptions of the likelihood of political instability and/or politically motivated violence, including terrorism.”
ID_GOVE_02	control of corruption	An indicator directly from the World Governance Indicators (WGI), “capturing perceptions from firms and households survey respondents and public, private, and NGO sector experts worldwide of public power exercised for private gain, including both petty and grand forms of corruption, as well as ‘capture’ of the state by elites and private interests.”
ID_GOVE_03	regulatory quality	An indicator directly from the World Governance Indicators (WGI), “capturing perceptions of the ability of the government to formulate and implement sound policies and regulations that permit and promote private sector development.”



ID_GOVE_04	rule of law	An indicator directly from the World Governance Indicators (WGI), “capturing perceptions from firms and households survey respondents and public, private, and NGO sector experts worldwide of confidence in and abide by the rules of society, and in particular the quality of contract enforcement, property rights, the police, and the courts, as well as the likelihood of crime and violence.”
ID_SOCI_03	education	A measure of enrolment in tertiary education to represent the education level of a country. It is approximated by the ratio of the enrollment in tertiary education (regardless of age) to the

SOCIAL READINESS

ID_SOCI_01	social inequality	The country’s poorest quintile’s share in national income or consumption.
ID_SOCI_02	ICT infrastructure	A composite indicator from 4 sub-indicators that consider both the access to and the use of ICT infrastructure: mobile phone subscription per 100 persons, fixed phone subscription per 100 persons, fixed broad-band subscription per 100 persons, and percent of individuals using the internet. Data for all four are available from the annual ICT Development Index (IDI) database. The mobile phone subscription measures the subscription to public mobile services including the post-paid and prepaid subscriptions (World Development Indicators, 2014). The fixed phone subscription is assumed to measure the active number of analog fixed telephone lines, ISDN channels, fixed wireless (WLL), public payphones and VoIP subscription (International Telecommunication Union, 2010). The fixed broad-band subscription refers to the number of broadband subscribers with a digital subscriber line, cable modem, or other high-speed technology (World Development Indicators, 2014). The individual internet use measures the proportion of internet users with access to the worldwide network (World Development Indicators, 2014).
ID_SOCI_03	education	A measure of enrollment in tertiary education to represent the education level of a country. It is approximated by the ratio of the enrollment in tertiary education (regardless of age) to the population of the age group that officially corresponds to tertiary education attendance.
ID_SOCI_04	innovation	A measure of the number of patent applications, filed through the Patent Cooperation Treaty procedure or with a national patent office, by residents per capita.



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COUNTRY INDEX CODEBOOK